

The Engineering Behind Modern Databases: B-Trees, Wal, Mvcc And Distributed Storage

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Half-title and publisher imprint.

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SYSTEM PRINCIPLE

Deterministic state transitions ensure reproducibility across replicas regardless of message arrival interleaving.

Operational telemetry and structured logging provide necessary observability into replica drift and compaction latency. Sustained reliability requires automated partition detection, quorum health monitoring, and self-healing node replacement workflows. See PostgreSQL Documentation for reference implementation details.

PUBLICATION INFORMATION

The Engineering Behind Modern Databases: B-Trees, Wal, Mvcc And Distributed Storage

A Definitive Technical Deep Dive

PUBLISHED BY

VasukiSquare AI Publishing Engine

EDITION

First Edition (2026)

RIGHTS HOLDER

VasukiSquare Technical Publishing

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Historical Context & Evolution

The evolution of data architectures reflects shifting hardware trade-offs from disk spindle contention to network fabric latency.

Table 1.2: Chronological progression of distributed storage architectures.

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Early Era	Single-Node ACID Engines	B-Tree indices & ARIES recovery protocol
Scaling Era	Distributed Key-Value Stores	Consistent hashing & Dynamo replication
Modern Era	NewSQL Distributed RDBMS	TrueTime / Raft consensus & Spanner transactions
Next Gen	Serverless & Memory-Tiered	NVMe-over-Fabrics & disaggregated compute/storage

Historical analysis of data storage paradigms 1970–2026.

Contemporary systems increasingly leverage disaggregated compute and storage, offloading replication protocols to hardware-accelerated interconnects.

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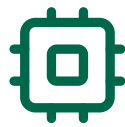
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CHAPTER 2

Architectural Principles

System Topology & Component Model

The write path ensures zero data loss by recording mutations to disk before acknowledging client requests, while maintaining high throughput through tiered background compaction.

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graph TD
    Client[Client Write] -->|1. Write| WAL[(Write-Ahead Log)]
    Client -->|2. Insert| MemTable[In-Memory MemTable]
    MemTable -->|3. Flush Threshold| Immutable[Immutable MemTable]
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    L0 -->|5. Compaction| L1[Level 1 SSTables]
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Figure 1.1: LSM-Tree Ingestion & Compaction Pipeline

Background compaction continuously merges overlapping key intervals from Level 0 to Level 1, bounding the number of disk seeks required for point and range lookups.

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Practical Implementation

Reference Implementation Patterns

The consensus engine coordinates state replication across clustered nodes via **atomic term transitions**. Each cluster participant transitions between *Follower*, *Candidate*, and *Leader* states based on heartbeat timeouts and term progression.

consensus/raft.rs

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pub struct RaftNode {
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Listing 1.1: Raft election logic and term advancement in Rust.

IMPLEMENTATION DETAIL

Always persist `voted_for` and `current_term` to non-volatile WAL storage prior to acknowledging RPC vote requests to prevent split-brain quorum anomalies.

By guaranteeing monotonic term increases and strict leader completeness, log entries committed in prior terms remain immutable across future leadership transitions.

Integration & Developer Experience

Choosing an appropriate storage engine requires evaluating trade-offs between **write amplification**, **read latency**, and cache efficiency under production concurrency constraints.

Table 1.1: Architectural trade-offs between B+ Trees and LSM Trees.

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CHAPTER 4

Performance, Benchmarks & Trade-offs

Throughput & Latency Benchmarks

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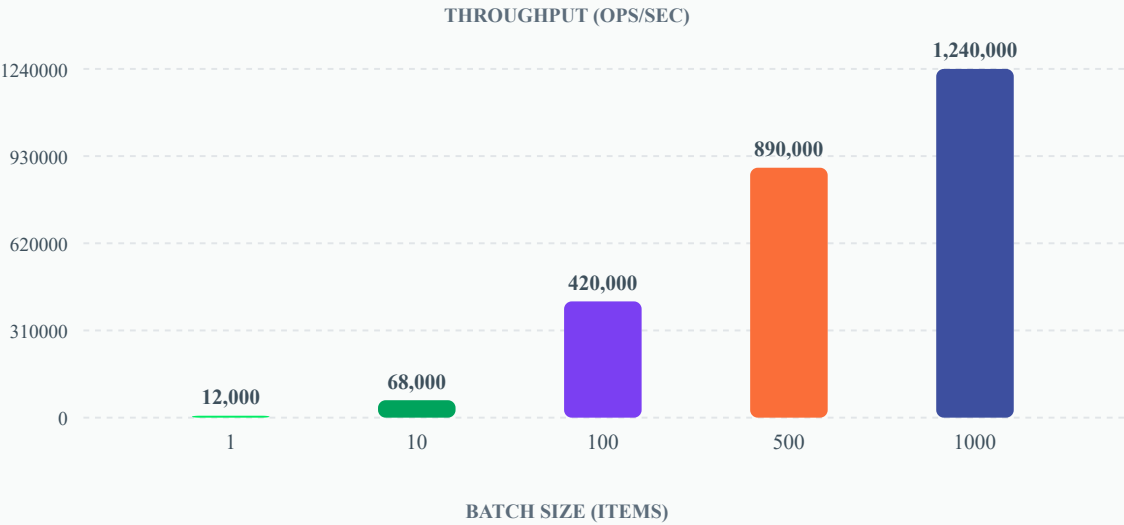
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OPERATIONS PER SECOND

Sustained write throughput benchmarked across a 64-node distributed NVMe cluster under 99.9th percentile SLA constraints.

Write Throughput Scaling by Batch Size

Empirical benchmark across 64 distributed storage nodes



Source: VasukiSquare Systems Research Benchmark Suite.

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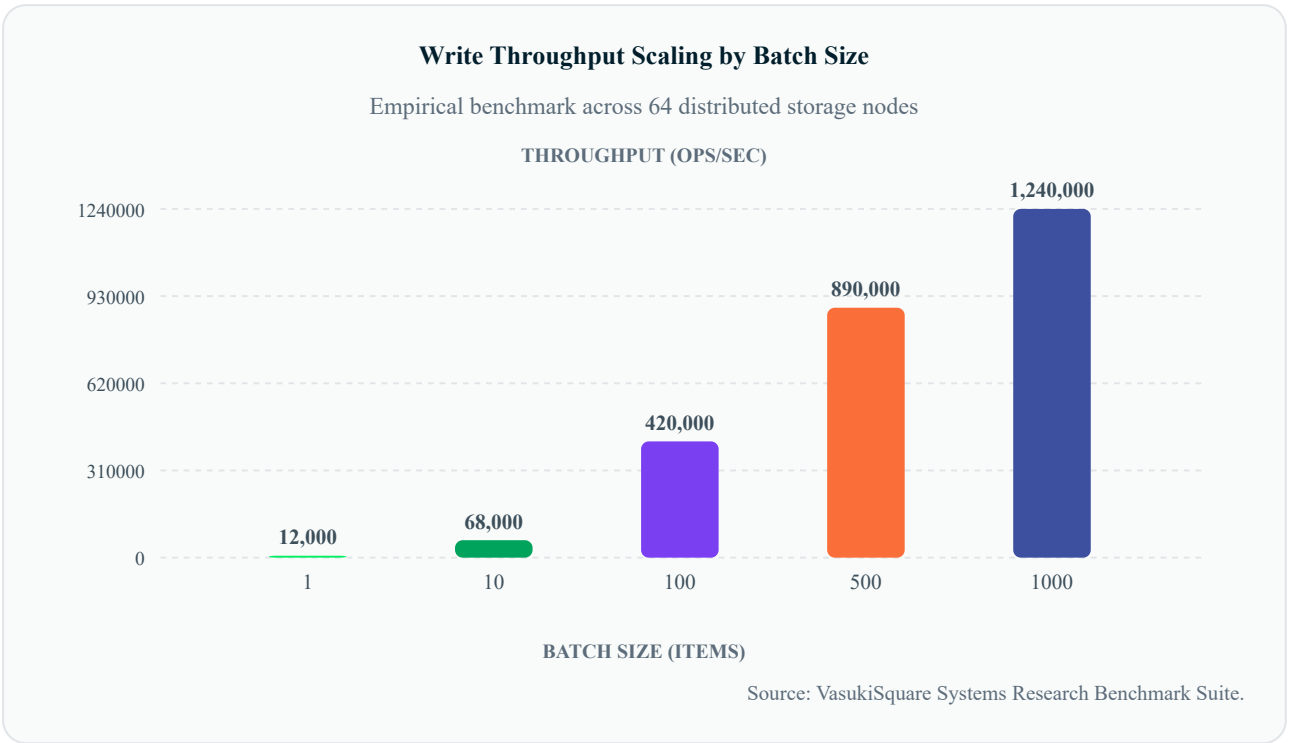
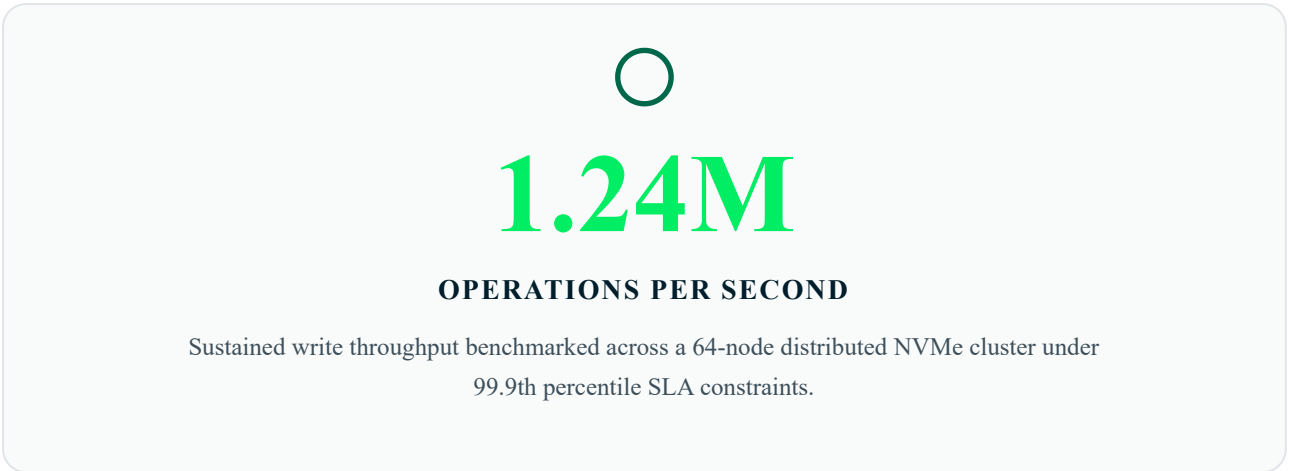
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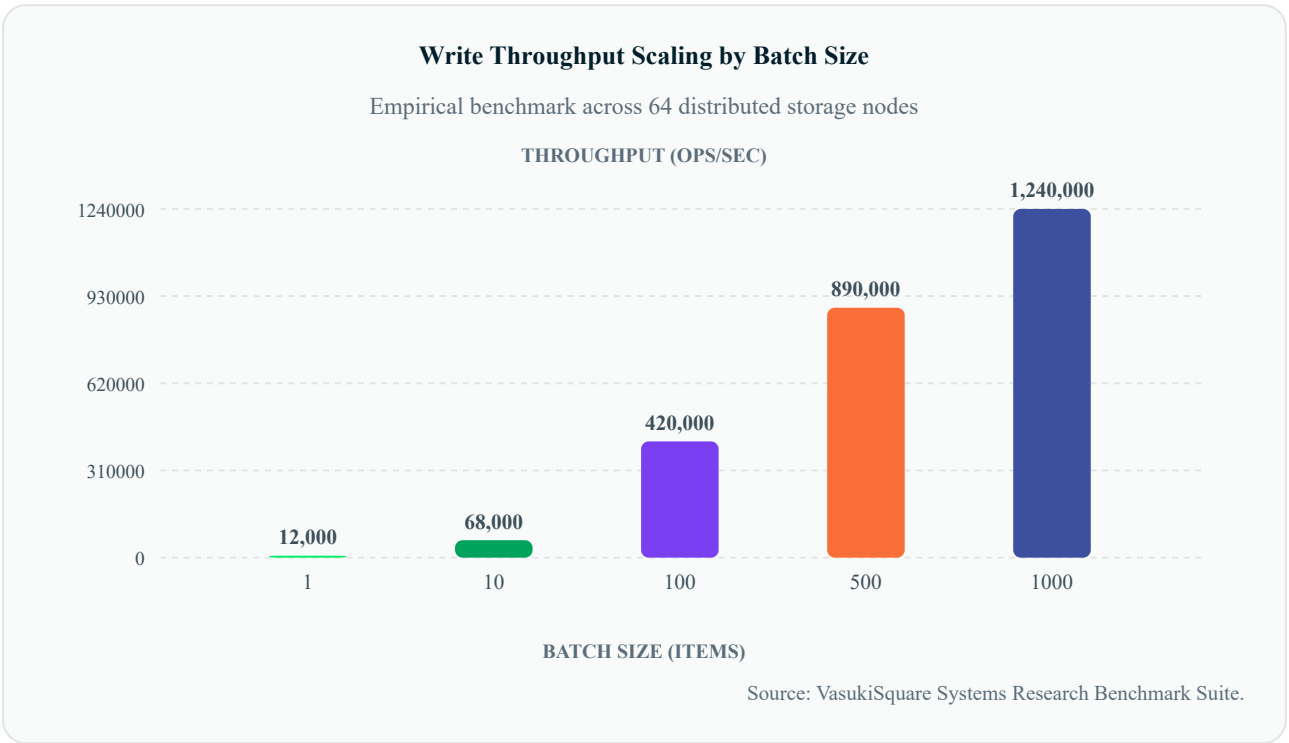
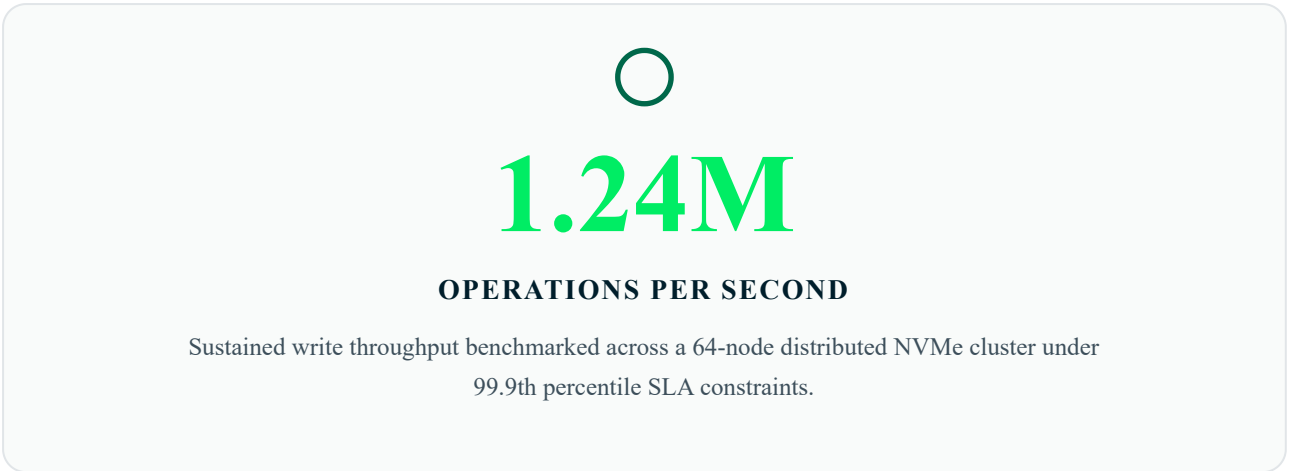
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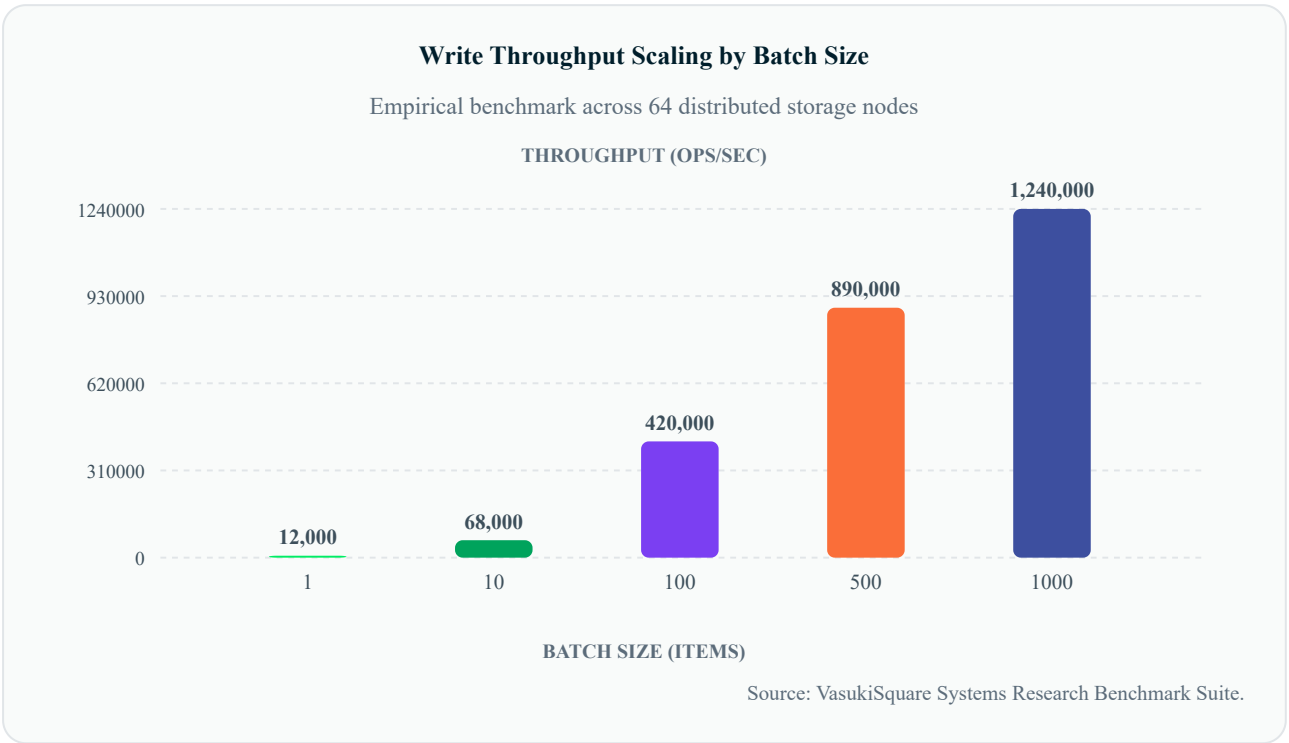
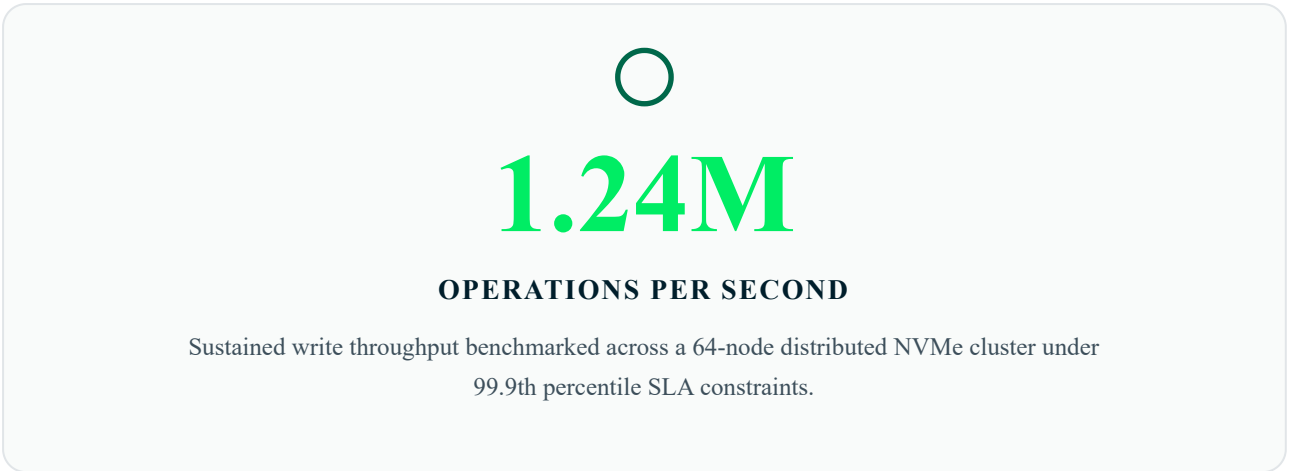
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Production Best Practices & Case Studies

Scaling & Reliability Strategies

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Simplicity is prerequisite for reliability. Complex recovery protocols inevitably create unforeseen failure modes.

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When building large-scale distributed systems, choosing deterministic, well-understood protocols dramatically simplifies operational troubleshooting and post-incident analysis.



KEY TAKEAWAY

Favor explicit state transitions and bounded queues over speculative buffering and unbounded retry policies.

Production Case Study & Retrospective

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References & Primary Sources

RESEARCH CORPUS

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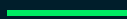
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Acknowledgements

Recognizing the open-source engineering foundations and academic research behind modern distributed data infrastructure.

This technical publication stands on the collective contributions of distributed systems researchers, database architects, and the global open-source community. Special gratitude is extended to the authors of the Raft consensus specification, PostgreSQL internals documentation, LSM-Tree storage architectures, and modern memory-tiered database systems. Their dedication to open scholarship and rigorous verification makes high-reliability technical publishing possible.



THANK YOU

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